

Related Work: Step-Level Feedback Distillation

1 Related Work

1.1 Process supervision and process reward models

A line of work has established that supervising how a model reasons, step by step is more effective than supervising only whether the final answer is correct. Lightman et al (1) compared the two regimes directly and found that process supervision substantially outperforms outcome supervision on the challenge MATH dataset, releasing the large PRM800K corpus of step level human labels to support the finding. Because human step annotation is expensive. Math-Shepherd (2) showed that process supervision can be constructed automatically, which is assigning a reward to each solution step without manual labels and used the resulting signal both to rerank candidate solutions and to drive step-level reinforcement learning. These process reward models (PRMs) treat each step’s correctness as a **scalar** quantity and deploy the PRM as a frozen scorer or ranker. Our work shares the step level granularity but departs from the scalar view: we treat the teacher step judgment, including its natural language critique, as the object to be distilled rather than a number to be predicted.

1.2 Generative PRMs

A more recent thread makes the verifier itself generative, producing a reasoning trace before it judges. GenPRMs(3) performs explicit chain-of-thought reasoning with code verification before scoring each step, while ThinkPRM(4) fine-tunes a long-COT verifier that articulates a verification rationale for every step and reaches strong ProcessBench performance from only a few thousand process labels. These models are close to ours in spirit they generate critiques rather than emit bare scores - but they are designed as verifiers evaluated at inference time, not as teacher whose score-plus-critique behavior is distilled into a separate, smaller, answer-blind student. Crucially, none of them isolates when privileged (answer-aware) judgment actually helps, which the question we take up.

1.3 Expert–amateur contrast

Our framing descends from work that extracts signal from the gap between a stronger and a weaker model. CLEAR (5) contrasts feedback from a larger "expert" and a smaller "amateur" model and applies the refined difference to iteratively improve outputs across reasoning tasks; it is the direct predecessor of the present project. Light-Reasoner (6) similarly exploits the behavioral divergence between a strong expert and a weak amateur to pinpoint high-value reasoning moments and distill them, which is notably without ground-truth labels, to sharpen reasoning. We retain the ground-truth free constraints at student test time but invert the asymmetry of interest rather than contrasting models of different capacity. We study a teacher with a privileged view, which has access to reference solution while labeling, distilled into a student that is **answer-blind** by construction. The relevant axis for us is information, not size.

1.4 Our position

Against this backdrop, we make two contributions, and we are deliberate about which is load-bearing. We characterize *when* privileged supervision is worth anything and report privilege $\times$ difficulty $\times$ richness effect: a teacher’s answer-aware judgment improves step-error localization only when the problem is hard enough that the teacher cannot self-verify *and* the privilege is rich enough to act on. On easy problems (GSM8K) a capable teacher already self-verifies and the privilege gap collapses to roughly zero. On hard-but-tractable problems (MATH) a bare final answer still moves nothing, while a full worked reference solution lifts error recall by a clear, statistically significant margin; on the hardest tier (OlympiadBench) the reference becomes too long to exploit and the gap closes again. To our knowledge this conditional structure, privilege buys signal precisely where self-verification fails, and only when the privileged information is a reasoning trace rather than a number, which has not been isolated in the PRM or expert-amateur literature. This is the validated result and the paper’s spine.

2 Results

2.1 Setup and metric

We evaluate a teacher’s ability to localize the first **erroneous step** in a multi-step solution under three supervision conditions that differ only in the privileged information made available to the teacher at labeling time: (i) **no-GT**, an answer-blind condition in which the teacher sees only the problem and the candidate solution; (ii) **+answer**, in which the teacher is additionally given in the final reference answer; and (iii) **solution**, in which the teacher is given the full worked reference solution. Performance is reported as Process Bench-style first-error **F1**, together with error-detection accuracy. The student remains ground-truth-free at test time throughout; privilege is a property of

the teacher’s labeling pass, not of the deployed evaluator. Reported deltas are taken against the no-GT condition, which serves as the unprivileged reference point.

2.2 Teacher selection

Before probing the effect of privilege, we fix the teacher by an upper-bound bake-off in which every candidate is granted ground-truth access (the with-GT ceiling, $N=30$). The official Gemma-class checkpoint attains the strongest first-error F_1 (0.909) with perfect format compliance (zero parse failures) and the highest throughput among the candidates; a comparably-sized Qwen variant follows closely (F_1 0.873). Notably, a reasoning-tuned candidate collapses under the structured-output protocol, failing to parse on the large majority of instances (parse-failure rate 0.846, F_1 0.000). This is a reminder that verbal reasoning fluency does not by itself confer reliable structured step adjudication. We therefore adopt the Gemma checkpoint as the primary teacher and retain the Qwen variant for the cross-family replication reported below

Teacher	F_1	correct_acc	error_acc	parse_fail
Gemma-4-26b (primary)	0.909	1.000	0.833	0.000
Qwen-27B	0.873	0.917	0.833	0.000
Qwen-35B-A3B	0.839	1.000	0.722	0.019
Qwen-4B-Thinking	0.000	1.000	0.000	0.846

Table 1: Teacher bake-off, with-GT ceiling ($N = 30$).

2.3 The privilege sweet spot

Our central finding is that privileged supervision is uniformly beneficial: it helps only when the problem is hard enough that the teacher cannot self-verify and tractable enough that the teacher can still exploit the reference. The solution privilege gap is non-significant on easy problems (GSM8K, -0.05 , n.s.) opens to a significant $+0.05$ F_1 on hard-but-tractable problems (MATH, 95% CI $[0.01, 0.09]$), and is non-significant again on the hardest tier (OlympiadBench, -0.03 , n.s.). The three points trace an inverted-U in difficulty, but we do not lean on the two non-significant endpoints to assert a curve; the curvature is established by mechanism in §??, where the rescue rate declines monotonically with reference length. The reading the data support is conjunctive: privilege pays off only where the teacher both *needs* the reference and can *use* it.

Table 2 reports the three tiers in full, together, with the Qwen-27B cross-family replication. On GSM8K, neither the bare answer nor the full solution improves an already-saturated judgment; the teacher is, in effect, its own oracle, and external privilege is redundant. On MATH, where unaided self-verification degrades, the full worked solution lifts first-error F_1 from 0.726 to 0.777 ($+0.05$, 95% CI $[0.01, 0.09]$, significant) and error recall from 0.608 to 0.661. The smaller preliminary MATH run ($N=50$) points the same way with a larger nominal gap ($\sim +0.13$ F_1). OlympiadBench, harder still, returns the gap to ≈ 0 : there the references are too long for the teacher to follow, so the privilege it is handed is no longer usable. The same inverted-U faintly reappears within MATH by difficulty level (strongest near the middle, weakest at the easiest and hardest levels), corroborating that tractability is the moderator; but the load-bearing evidence is the three-tier contrast, not the noisier per-level split. Critically, the cross-family Qwen-27B teacher reproduces the effect on MATH ($+0.082$ F_1 , error detection $0.526 \rightarrow 0.667$), which substantially weakens any single-model-artifact explanation.

Tier	Benchmark (teacher)	no-GT	+answer	+solution	Δ (solution)
Easy	GSM8K (Gemma, $N=50$)	0.889	0.856	0.837	-0.052 (n.s.)
Hard	MATH (Gemma, $N=400$)	0.726	0.726	0.777	$+0.051$ $[0.01, 0.09]$
Hardest	OlympiadBench (Gemma, OE)	—	—	—	-0.03 (n.s.)
Hard	MATH (Qwen-27B, $N=150$)	0.653	0.690	0.735	$+0.082$

Table 2: The privilege sweet spot across difficulty tiers, with cross-family replication. Values are first-error F_1 . The solution gap rises then falls—an inverted-U, not a monotone trend—and reproduces across two model families. MATH (Gemma) is the $N=400$ definitive run (95% CI shown); OlympiadBench is OE-only, verified.

2.4 Richness, not mere access

The MATH panel isolates a second, sharper effect. The +ANSWER condition, which is a correct final answer but no reasoning, is indistinguishable from the answer-blind baseline: the two rows are identical and the bare answer flips *zero* predictions (measured at $N=150$; we verified the labeling pipeline is correctly wired, so the null effect is behavioral, but not artifact). Only the full worked solution moves the metric. The operative variable is therefore not whether the teacher is privileged but whether the privilege is *rich*: a reference reasoning trace supplies the step-by-step scaffold

needed to localize an error, whereas a final answer, which the teacher cannot back-propagate into the faulty step, supplies nothing it can act on.

2.5 Mechanism: privilege rescues self-verification failures

The sweet spot is not a raw difficulty effect; it is a *rescue of self verification failures*, and decomposes into two gates ($N=400$). **Gate 1 (the teacher must need help)**. Of 227 problems containing an error, the NO-GT teacher misses 89. Adding the worked solution rescues 29 of those (33%), while breaking only 17 of the 138 it had already caught (12%), which is the benefit concentrates exactly where self-verification fails. **Gate 2(the teacher must be able to use it)**. The rescue rate fails monotonically with the length of the reference solution (short 0.37 $\rightarrow$ mid 0.33 $\rightarrow$ long 0.28), which is the mechanism behind the collapse at the hardest tier. Together the gates identify the moderator as self-verification-failure $\times$ tractability, precisely the inverted-U observed across and within datasets.

Error type	Why no-GT passed it	Why +solution flagged it
Unjustified algebraic step	[step] — the manipulation looked locally valid	[the reference takes a different, justified route]
Sign / arithmetic slip	[step] — masked by a plausible continuation	[the reference’s value diverges at that step]
Invalid case split	[step] — each branch seemed self-consistent	[the reference enumerates the complete case set]

Table 3: Representative flip cases by error type (2–3 to be finalized from the evidence pack).

2.6 Does the teacher advantage transfer to the student? (a verified negative)

The teacher-level sweet spot above is the validated result. The natural next question is whether it *distills*: if we train a small (1.5B) student PRM on the privileged teacher’s score-plus-critique labels, does that student inherit a usable advantage over a student trained on no-GT labels? Across a validated $N=1000$ run (1,000 training problems, 400 evaluation problems, Gemma-4 labeling confirmed through $\sim 32k$ served requests), **the answer is no**. We report this as a clean negative.

Step-level transfer. We compare three students, which are privileged + critique, privileged score-only, and no-GT+critique, on threshold-free ROC-AUC, the appropriate measure here because first error F_1 at a fixed score-head cutoff moves with calibration rather than capability. Table 4 shows the no-GT student is the *highest* on ROC-AUC (0.641 vs .631), and the run is non-degenerate (predicted-error rate ≈ 0.3 across all cells, so no silent collapse inflates the comparison). Privilege does not produce a better student verifier.

Student (teacher labels)	ROC-AUC	PR-AUC	F_1	error_recall
Privileged + critique	0.631	0.130	0.170	0.357
Privileged, score-only	0.624	0.135	0.184	0.427
No-GT + critique	0.641	0.121	0.198	0.467

Table 4: Step-level transfer ($N_{\text{train}}=1000$, $N_{\text{eval}}=400$). Compared on ROC-AUC, the no-GT student is highest—privilege does not transfer.

2.7 Downstream best-of- N

The motivating promise of a ground-truth-free step-error localizer is a test-time gain: give N sampled solutions, the student PRM scores each and we re-rank, selecting the highest-score candidate (best-of- N). Table 5 reports the $N=1000$ result. The no-GT verifier actually re-ranks *better* than the privileged one (0.373 vs..349), and decisively **neither beats the majority-vote baseline** (≈ 0.39). The oracle (0.517) shows substantial headroom remains, so the failure is the verifiers, not the candidate pool’s.

Verifier	pass@1	prm_rerank	majority_vote	oracle_pass@N
No-GT student	0.337	0.373	0.391	0.517
Privileged student	0.335	0.349	0.382	0.518

Table 5: Downstream best-of- N re-ranking ($N=1000$, $N=8$ candidates, **gemma-2-9b-it** generator). The no-GT verifier re-ranks better; neither beats majority vote.

Why the null, such as a diffuse, not absent, teacher gap. The negative is not that privileged and no-GT labels are identical. A probe of the actual Gemma-4 labeling teacher reproduces a real +0.07 solution gap (F_1 0.716 $\rightarrow$ 0.786; error recall 0.577 $\rightarrow$ 0.654, $N=150$ MATH). The issue is the *shape* of the gap. Over 6, 340 labeled steps, privileged and no-GT labels agree on only 69%, which they differ on $\sim 31\%$ of steps, but the disagreement is nearly symmetric: privilege flags an error where no-GT does not on 1,001 steps, while no-GT flags one where privilege does not on 956. The teacher’s privilege churns labels in *both* directions; its small net quality gain is *diffuse*, not a clean directional signal a 1.5B student can latch onto. A same-pool paired best-of- N confirms the consequence: the privileged and no-GT re-rankers are statistically indistinguishable (McNemar $p=0.14$; 5 vs. 12 discordant pairs), both below majority vote. The honest summary is mechanistic: *a real teacher-quality does not distill because it is diffuse*.

Boundary of the claim. Two caveats keep this null appropriately scoped, and define the forward work. First, the paired test at $N=200$ is underpowered (17 discordant pairs); a re-run at $N=1000$ is queued. Second, and more important, both students currently lose to majority vote, so strictly the present result is that privilege does not distill into a *weak* verifier. Establishing the boundary cleanly therefore requires (i) a student that first beats majority vote (data scale, larger base), and (ii) a strong-vs-weak-teacher positive control confirming the pipeline can detect transfer when it genuinely exists. These turn a flat null into a publishable transfer-boundary result.

References

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